

Supplementary Material for “Inner c -Differential Cryptanalysis of Kuznyechik”

Pantelimon Stănică^a, Ranit Dutta^b, Bimal Mandal^b

^a*Applied Mathematics Department, Naval Postgraduate School, Monterey, CA 93943, USA*

^b*Department of Mathematics, Indian Institute of Technology Jodhpur, Jodhpur, Rajasthan 342030, India*

Abstract

This supplementary document contains completeness and reproducibility material for the accompanying article: the full Kuznyechik technical specification used in the computations, complete inner/outer c -differential-uniformity tables, detailed round-by-round exploratory cryptanalysis results, and a representative statistical-analysis output. The main article contains the definitions, structural theorems, proofs, exact two-round key-recovery analysis, and principal cryptanalytic conclusions.

Appendix A. Kuznyechik Block Cipher: Technical Specification: GOST R 34.12-2015

We include the full Kuznyechik specification here for reproducibility and self-containment, as the original GOST R 34.12-2015 standard is in Russian. The specification follows RFC 7801.

Kuznyechik is a symmetric block cipher standardized as GOST R 34.12-2015, serving as the official encryption standard of the Russian Federation. The cipher operates on 128-bit blocks with a 256-bit key, employing a substitution-permutation network (SPN) structure over 9 rounds, plus a key whitening at the backend. The cipher operates over the finite field $\mathbb{F}_{2^8}$ with the irreducible polynomial $p(x) = x^8 + x^7 + x^6 + x + 1$. Each 128-bit block is treated as a vector of 16 bytes (elements of $\mathbb{F}_{2^8}$).

The S-box

The Kuznyechik S-box is a bijective mapping $S : \mathbb{F}_{2^8} \rightarrow \mathbb{F}_{2^8}$ designed to provide strong nonlinear properties. The S-box is defined by the following

16×16 lookup table (values in hexadecimal, row-major):

	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
0	0xfc	0xee	0xdd	0x11	0xcf	0x6e	0x31	0x16	0xfb	0xc4	0xfa	0xda	0x23	0xc5	0x04	0x4d
1	0xe9	0x77	0xf0	0xdb	0x93	0x2e	0x99	0xba	0x17	0x36	0xf1	0xbb	0x14	0xcd	0x5f	0xc1
2	0xf9	0x18	0x65	0x5a	0xe2	0x5c	0xef	0x21	0x81	0x1c	0x3c	0x42	0x8b	0x01	0x8e	0x4f
3	0x05	0x84	0x02	0xae	0xe3	0x6a	0x8f	0xa0	0x06	0x0b	0xed	0x98	0x7f	0xd4	0xd3	0x1f
4	0xeb	0x34	0x2c	0x51	0xea	0xc8	0x48	0xab	0xf2	0x2a	0x68	0xa2	0xfd	0x3a	0xce	0xcc
5	0xb5	0x70	0x0e	0x56	0x08	0x0c	0x76	0x12	0xbf	0x72	0x13	0x47	0x9c	0xb7	0x5d	0x87
6	0x15	0xa1	0x96	0x29	0x10	0x7b	0x9a	0xc7	0xf3	0x91	0x78	0x6f	0x9d	0x9e	0xb2	0xb1
7	0x32	0x75	0x19	0x3d	0xff	0x35	0x8a	0x7e	0x6d	0x54	0xc6	0x80	0xc3	0xbd	0x0d	0x57
8	0xdf	0xf5	0x24	0xa9	0x3e	0xa8	0x43	0xc9	0xd7	0x79	0xd6	0xf6	0x7c	0x22	0xb9	0x03
9	0xe0	0x0f	0xec	0xde	0x7a	0x94	0xb0	0xbc	0xdc	0xe8	0x28	0x50	0x4e	0x33	0x0a	0x4a
A	0xa7	0x97	0x60	0x73	0x1e	0x00	0x62	0x44	0x1a	0xb8	0x38	0x82	0x64	0x9f	0x26	0x41
B	0xad	0x45	0x46	0x92	0x27	0x5e	0x55	0x2f	0x8c	0xa3	0xa5	0x7d	0x69	0xd5	0x95	0x3b
C	0x07	0x58	0xb3	0x40	0x86	0xac	0x1d	0xf7	0x30	0x37	0x6b	0xe4	0x88	0xd9	0xe7	0x89
D	0xe1	0x1b	0x83	0x49	0x4c	0x3f	0xf8	0xfe	0x8d	0x53	0xaa	0x90	0xca	0xd8	0x85	0x61
E	0x20	0x71	0x67	0xa4	0x2d	0x2b	0x09	0x5b	0xcb	0x9b	0x25	0xd0	0xbe	0xe5	0x6c	0x52
F	0x59	0xa6	0x74	0xd2	0xe6	0xf4	0xb4	0xc0	0xd1	0x66	0xaf	0xc2	0x39	0x4b	0x63	0xb6

Table A.1: Kuznyechik S-box

The inverse S-box S^{-1} is computed as the multiplicative inverse of the S-box mapping. It is defined by the following 16×16 lookup table (values in hexadecimal, row-major):

	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
0	0xa5	0x2d	0x32	0x8f	0x0e	0x30	0x38	0xc0	0x54	0xe6	0x9e	0x39	0x55	0x7e	0x52	0x91
1	0x64	0x03	0x57	0x5a	0x1c	0x60	0x07	0x18	0x21	0x72	0xa8	0xd1	0x29	0xc6	0xa4	0x3f
2	0xe0	0x27	0x8d	0x0c	0x82	0xea	0xae	0xb4	0x9a	0x63	0x49	0xe5	0x42	0xe4	0x15	0xb7
3	0xc8	0x06	0x70	0x9d	0x41	0x75	0x19	0xc9	0xaa	0xfc	0x4d	0xbf	0x2a	0x73	0x84	0xd5
4	0xc3	0xaf	0x2b	0x86	0xa7	0xb1	0xb2	0x5b	0x46	0xd3	0x9f	0xfd	0xd4	0x0f	0x9c	0x2f
5	0x9b	0x43	0xef	0xd9	0x79	0xb6	0x53	0x7f	0xc1	0xf0	0x23	0xe7	0x25	0x5e	0xb5	0x1e
6	0xa2	0xdf	0xa6	0xfe	0xac	0x22	0xf9	0xe2	0x4a	0xbc	0x35	0xca	0xee	0x78	0x05	0x6b
7	0x51	0xe1	0x59	0xa3	0xf2	0x71	0x56	0x11	0x6a	0x89	0x94	0x65	0x8c	0xbb	0x77	0x3c
8	0x7b	0x28	0xab	0xd2	0x31	0xde	0xc4	0x5f	0xcc	0xcf	0x76	0x2c	0xb8	0xd8	0x2e	0x36
9	0xdb	0x69	0xb3	0x14	0x95	0xbe	0x62	0xa1	0x3b	0x16	0x66	0xe9	0x5c	0x6c	0x6d	0xad
A	0x37	0x61	0x4b	0xb9	0xe3	0xba	0xf1	0xa0	0x85	0x83	0xda	0x47	0xc5	0xb0	0x33	0xfa
B	0x96	0x6f	0x6e	0xc2	0xf6	0x50	0xff	0x5d	0xa9	0x8e	0x17	0x1b	0x97	0x7d	0xec	0x58
C	0xf7	0x1f	0xfb	0x7c	0x09	0x0d	0x7a	0x67	0x45	0x87	0xdc	0xe8	0x4f	0x1d	0x4e	0x04
D	0xeb	0xf8	0xf3	0x3e	0x3d	0xbd	0x8a	0x88	0xdd	0xcd	0x0b	0x13	0x98	0x02	0x93	0x80
E	0x90	0xd0	0x24	0x34	0xcb	0xed	0xf4	0xce	0x99	0x10	0x44	0x40	0x92	0x3a	0x01	0x26
F	0x12	0x1a	0x48	0x68	0xf5	0x81	0x8b	0xc7	0xd6	0x20	0x0a	0x08	0x00	0x4c	0xd7	0x74

Table A.2: Kuznyechik Inverse S-box

Linear Transformation Layer (L -Layer)

The linear transformation L operates on 128-bit blocks, treating them as vectors of 16 bytes over $\mathbb{F}_{2^8}$. The Kuznyechik linear transformation is defined by applying a basic 1-byte cyclic shift with XOR feedback, called R , sixteen times. With the byte ordering used by the RFC-tested implementation in this

paper, define the feedback byte $\ell(\mathbf{a}) = \bigoplus_{j=0}^{15} c_j \cdot a_j$. The basic transformation is $R(a_0, a_1, \dots, a_{15}) = (\ell(\mathbf{a}), a_0, a_1, \dots, a_{14})$, where

$$\mathbf{c} = (0\text{x}94, 0\text{x}20, 0\text{x}85, 0\text{x}10, 0\text{x}c2, 0\text{x}c0, 0\text{x}01, 0\text{x}fb, \\ 0\text{x}01, 0\text{x}c0, 0\text{x}c2, 0\text{x}10, 0\text{x}85, 0\text{x}20, 0\text{x}94, 0\text{x}01).$$

The coefficients lie in $\mathbb{F}_{2^8}$. The full map $L : \mathbb{F}_{2^8}^{16} \rightarrow \mathbb{F}_{2^8}^{16}$ is $L(\mathbf{a}) = R^{16}(\mathbf{a})$. For efficient software implementation, the linear transformation $\mathbf{L}$ and its inverse $\mathbf{L}^{-1}$ are typically precomputed into lookup tables. The Python code utilizes such precomputed tables for fast execution.

Key Schedule

The key schedule transforms the 256-bit master key into ten 128-bit round keys $K^{(0)}, K^{(1)}, \dots, K^{(9)}$. The algorithm uses a Feistel-like structure with 32 iterations to generate intermediate key values, from which the round keys are derived. The key schedule employs iteration constants $\mathbf{C}_j$ for $j = 1, 2, \dots, 32$. With the byte ordering of this appendix, the iteration constant is $\mathbf{C}_j = L(0, \dots, 0, j)$, i.e., the integer j is placed in the final byte before applying L . The full list of these constants in hexadecimal (row-major for each 16-byte vector) is:

```

C1 = (0x6e, 0xa2, 0x69, 0x2a, 0xe9, 0x19, 0x37, 0x7e, 0xdc, 0xfe, 0x15, 0x41, 0x28, 0xd9, 0x21, 0x66)
C2 = (0xb7, 0x5f, 0x82, 0x79, 0x91, 0x13, 0xdd, 0x4c, 0xb5, 0x8f, 0x9f, 0x85, 0x1d, 0xad, 0x20, 0x69)
C3 = (0x90, 0x94, 0x4a, 0x96, 0x96, 0x48, 0x6c, 0xa6, 0x64, 0x2f, 0xce, 0x3b, 0x25, 0x21, 0x85, 0x86)
C4 = (0x96, 0x68, 0x8a, 0x92, 0x13, 0x11, 0xbf, 0x55, 0x02, 0x16, 0x48, 0x0a, 0x2c, 0x78, 0x68, 0x15)
C5 = (0x3a, 0x6c, 0x54, 0x74, 0x4b, 0x39, 0x52, 0x7c, 0xfb, 0x98, 0x45, 0x9c, 0xed, 0x38, 0xe4, 0x9e)
C6 = (0x39, 0x25, 0x23, 0x4a, 0x94, 0x11, 0x84, 0x57, 0x8e, 0x24, 0x69, 0x93, 0x0f, 0x8b, 0x73, 0x93)
C7 = (0xbc, 0x4f, 0x65, 0x76, 0x6e, 0x5f, 0x17, 0xdd, 0x03, 0x6f, 0x38, 0x07, 0x57, 0x24, 0x40, 0x90)
C8 = (0x97, 0x5c, 0x6f, 0x68, 0x11, 0xdd, 0x66, 0x0b, 0x59, 0x4a, 0x27, 0x15, 0x81, 0x3d, 0x76, 0x1c)
C9 = (0x2a, 0x2f, 0x56, 0x7b, 0x82, 0x48, 0x21, 0xa2, 0x22, 0xa2, 0x83, 0x09, 0x03, 0x1a, 0x4f, 0x07)
C10 = (0xa4, 0x6b, 0x5d, 0x26, 0x85, 0x33, 0x7f, 0x5e, 0x4b, 0x77, 0x47, 0x32, 0xa7, 0x17, 0x86, 0x42)
C11 = (0x85, 0x2c, 0x90, 0x2e, 0xa7, 0x03, 0x4f, 0x55, 0x91, 0x46, 0x77, 0x53, 0x56, 0x0b, 0x04, 0x32)
C12 = (0x7e, 0xb3, 0x92, 0x91, 0x33, 0x05, 0x27, 0xc5, 0x35, 0x27, 0x9e, 0x8b, 0x5b, 0x6e, 0x47, 0x84)
C13 = (0x90, 0xca, 0xd4, 0x21, 0x85, 0x05, 0x08, 0x47, 0x98, 0x85, 0x06, 0xb7, 0x0b, 0x0d, 0x17, 0x8f)
C14 = (0x94, 0x44, 0x3a, 0x81, 0x25, 0x0e, 0x96, 0x0b, 0x62, 0x56, 0x24, 0x68, 0x84, 0x01, 0x0d, 0x06)
C15 = (0xa1, 0x05, 0x66, 0x14, 0x78, 0xbd, 0x52, 0xdb, 0x42, 0x33, 0x0c, 0x0d, 0x5e, 0x20, 0x15, 0x0f)
C16 = (0x40, 0x21, 0x05, 0x05, 0xb3, 0x73, 0x8c, 0xf5, 0x5f, 0x57, 0x09, 0x36, 0x98, 0x31, 0x31, 0x73)
C17 = (0x2f, 0x33, 0x87, 0x86, 0x67, 0x03, 0x05, 0x32, 0x92, 0xf8, 0xbc, 0x41, 0x13, 0x06, 0x95, 0x26)
C18 = (0x6c, 0xa4, 0x9f, 0xf0, 0xb9, 0x54, 0x8a, 0x18, 0x06, 0x05, 0xa6, 0x85, 0x47, 0x65, 0x68, 0x68)
C19 = (0x51, 0x9c, 0x24, 0x49, 0x28, 0x69, 0x96, 0x26, 0x87, 0x35, 0x89, 0x67, 0x04, 0x91, 0x99, 0xa0)
C20 = (0x92, 0x82, 0x11, 0x87, 0x1a, 0x03, 0x59, 0x0e, 0x63, 0x26, 0x0a, 0x7f, 0xec, 0x22, 0x3a, 0x92)
C21 = (0x81, 0xb3, 0xc5, 0x12, 0x74, 0x26, 0x0e, 0x93, 0x6a, 0x66, 0x7f, 0xdc, 0x94, 0x58, 0x7d, 0x13)
C22 = (0x1a, 0x35, 0x9e, 0xd6, 0x28, 0x68, 0x21, 0x9b, 0x92, 0x05, 0x7c, 0x88, 0x81, 0xb3, 0xc5, 0x12)
C23 = (0x74, 0x26, 0x2e, 0x93, 0x6a, 0x66, 0x7f, 0xdc, 0x94, 0x58, 0x7d, 0x13, 0x75, 0x61, 0xba, 0x13)
C24 = (0x99, 0x34, 0x96, 0x88, 0x23, 0x9b, 0x56, 0x16, 0x80, 0x91, 0x4e, 0xfc, 0x22, 0x2d, 0x3a, 0x92)
C25 = (0x69, 0x85, 0x28, 0x83, 0x98, 0x47, 0x28, 0x69, 0x96, 0x26, 0x87, 0x35, 0x89, 0x67, 0x04, 0x91)
C26 = (0x99, 0xa0, 0x92, 0x82, 0x11, 0x87, 0x1a, 0x35, 0x9e, 0xd6, 0x28, 0x68, 0x21, 0x9b, 0x92, 0x05)
C27 = (0x67, 0x8b, 0x81, 0xb3, 0xc5, 0x12, 0x74, 0x26, 0x2e, 0x93, 0x6a, 0x66, 0x7f, 0xdc, 0x94, 0x58)
C28 = (0x7d, 0x13, 0x75, 0x61, 0xba, 0x13, 0x74, 0x26, 0x2e, 0x93, 0x6a, 0x66, 0x7f, 0xdc, 0x94, 0x58)
C29 = (0x7d, 0x13, 0x99, 0x34, 0x96, 0x88, 0x23, 0x9b, 0x56, 0x16, 0x80, 0x91, 0x4e, 0xfc, 0x22, 0x2d)
C30 = (0x3a, 0x92, 0x69, 0x85, 0x28, 0x83, 0x98, 0x47, 0x28, 0x69, 0x96, 0x26, 0x87, 0x35, 0x89, 0x67)
C31 = (0x04, 0x91, 0x99, 0xa0, 0x92, 0x82, 0x11, 0x87, 0x1a, 0x35, 0x9e, 0xd6, 0x28, 0x68, 0x21, 0x9b)
C32 = (0x92, 0x05, 0x67, 0x8b, 0x81, 0xb3, 0xc5, 0x12, 0x74, 0x26, 0x2e, 0x93, 0x6a, 0x66, 0x7f, 0xdc)

```

Key Schedule Algorithm

Algorithm 1 Kuznyechik Key Schedule (RFC-compatible indexing)

Require: 256-bit master key $K = \mathbf{K}_1 \parallel \mathbf{K}_2$, $|\mathbf{K}_1| = |\mathbf{K}_2| = 128$

Ensure: Round keys $K^{(0)}, \dots, K^{(9)}$

```

1:  $A \leftarrow \mathbf{K}_1, B \leftarrow \mathbf{K}_2$ 
2:  $K^{(0)} \leftarrow A, K^{(1)} \leftarrow B$ 
3: for  $j = 1$  to 32 do
4:    $T \leftarrow L(S(A \oplus \mathbf{C}_j)) \oplus B$ 
5:    $B \leftarrow A$ 
6:    $A \leftarrow T$ 
7:   if  $j \bmod 8 = 0$  then
8:      $r \leftarrow j/8$ 
9:      $K^{(2r)} \leftarrow A$ 
10:     $K^{(2r+1)} \leftarrow B$ 
11:   end if
12: end for

```

Encryption Algorithm

The encryption process applies 9 rounds of transformation followed by a final key addition:

Algorithm 2 Kuznyechik Encryption

```

1: Input: 128-bit plaintext  $\mathbf{P}$ , round keys  $K^{(0)}, \dots, K^{(9)}$ 
2: Output: 128-bit ciphertext  $\mathbf{C}$ 
3:  $\mathbf{X} \leftarrow \mathbf{P}$ 
4: for  $i = 0$  to 8 do
5:    $\mathbf{X} \leftarrow \mathbf{X} \oplus K^{(i)}$  AddRoundKey
6:    $\mathbf{X} \leftarrow S(\mathbf{X})$  SubBytes (byte-wise application of S-box)
7:    $\mathbf{X} \leftarrow L(\mathbf{X})$  Linear transformation
8: end for
9:  $\mathbf{C} \leftarrow \mathbf{X} \oplus K^{(9)}$  Final key addition return  $\mathbf{C}$ 

```

For completeness, we include the decryption algorithm, though our analysis uses only encryption. Decryption applies the inverse operations in reverse order:

Algorithm 3 Kuznyechik Decryption

```
1: Input: 128-bit ciphertext  $\mathbf{C}$ , round keys  $K^{(0)}, \dots, K^{(9)}$ 
2: Output: 128-bit plaintext  $\mathbf{P}$ 
3:  $\mathbf{X} \leftarrow \mathbf{C}$ 
4:  $\mathbf{X} \leftarrow \mathbf{X} \oplus K^{(9)}$  Remove final key
5: for  $i = 8$  downto 0 do
6:    $\mathbf{X} \leftarrow \mathbf{L}^{-1}(\mathbf{X})$  Inverse linear transformation
7:    $\mathbf{X} \leftarrow S^{-1}(\mathbf{X})$  Inverse SubBytes (byte-wise application of  $S^{-1}$ )
8:    $\mathbf{X} \leftarrow \mathbf{X} \oplus K^{(i)}$  Remove round key
9: end for
10:  $\mathbf{P} \leftarrow \mathbf{X}$ 
11: return  $\mathbf{P}$ 
```

Cryptanalytic context

The specification above is included for reproducibility rather than as a security claim. The Kuznyechik S-box has ordinary differential uniformity 8, so its largest nonzero-input DDT entry is $8/256 = 2^{-5}$. The L -layer has branch number 17, which is the diffusion fact used in the exact two-round channel argument of the main article. Kiryukhin computed the exact classical two-round MEDP $2^{-86.66\dots}$ [20, 21], while other literature studies reduced-round meet-in-the-middle, differential, algebraic, and integral properties [24, 25]. The reverse-engineering work on the S-box [7, 8, 9] is particularly relevant here because our inner- c results exploit structure that is not captured by the ordinary differential uniformity alone.

Appendix B. The (inner/outer) cDU Table for the S-box of Kuznyechik and its inverse

Recall the complementary property of the inner cDU of an S-box with the outer cDU of its inverse. For the purpose of completion, we include both here.

Table B.3: Inner c -Differential Uniformity δ_c of Kuznyechik S-box

c	Hex	δ_c	c	Hex	δ_c	c	Hex	δ_c	c	Hex	δ_c	c	Hex	δ_c
1	0x01	8	52	0x34	11	103	0x67	8	154	0x9a	7	205	0xcd	7
2	0x02	64	53	0x35	13	104	0x68	8	155	0x9b	7	206	0xce	8
3	0x03	21	54	0x36	8	105	0x69	8	156	0x9c	8	207	0xcf	7
4	0x04	33	55	0x37	7	106	0x6a	9	157	0x9d	12	208	0xd0	7
5	0x05	9	56	0x38	8	107	0x6b	9	158	0x9e	11	209	0xd1	8
6	0x06	10	57	0x39	8	108	0x6c	7	159	0x9f	8	210	0xd2	7
7	0x07	8	58	0x3a	9	109	0x6d	6	160	0xa0	7	211	0xd3	7
8	0x08	13	59	0x3b	11	110	0x6e	6	161	0xa1	7	212	0xd4	8

9	0x09	8	60	0x3c	8	111	0x6f	13	162	0xa2	7	213	0xd5	8
10	0x0a	13	61	0x3d	7	112	0x70	10	163	0xa3	7	214	0xd6	8
11	0x0b	11	62	0x3e	14	113	0x71	9	164	0xa4	8	215	0xd7	7
12	0x0c	8	63	0x3f	14	114	0x72	9	165	0xa5	9	216	0xd8	8
13	0x0d	7	64	0x40	8	115	0x73	8	166	0xa6	9	217	0xd9	8
14	0x0e	7	65	0x41	9	116	0x74	9	167	0xa7	8	218	0xda	7
15	0x0f	8	66	0x42	7	117	0x75	12	168	0xa8	8	219	0xdb	8
16	0x10	11	67	0x43	7	118	0x76	9	169	0xa9	13	220	0xdc	7
17	0x11	7	68	0x44	7	119	0x77	8	170	0xaa	7	221	0xdd	7
18	0x12	8	69	0x45	7	120	0x78	9	171	0xab	8	222	0xde	8
19	0x13	7	70	0x46	7	121	0x79	9	172	0xac	7	223	0xdf	8
20	0x14	9	71	0x47	8	122	0x7a	7	173	0xad	7	224	0xe0	13
21	0x15	9	72	0x48	8	123	0x7b	8	174	0xae	8	225	0xe1	64
22	0x16	7	73	0x49	7	124	0x7c	8	175	0xaf	9	226	0xe2	7
23	0x17	8	74	0x4a	8	125	0x7d	7	176	0xb0	8	227	0xe3	8
24	0x18	8	75	0x4b	8	126	0x7e	8	177	0xb1	8	228	0xe4	8
25	0x19	7	76	0x4c	8	127	0x7f	8	178	0xb2	7	229	0xe5	7
26	0x1a	7	77	0x4d	8	128	0x80	9	179	0xb3	8	230	0xe6	8
27	0x1b	8	78	0x4e	8	129	0x81	8	180	0xb4	7	231	0xe7	9
28	0x1c	8	79	0x4f	7	130	0x82	8	181	0xb5	11	232	0xe8	8
29	0x1d	12	80	0x50	8	131	0x83	8	182	0xb6	8	233	0xe9	8
30	0x1e	7	81	0x51	8	132	0x84	7	183	0xb7	7	234	0xea	9
31	0x1f	8	82	0x52	7	133	0x85	8	184	0xb8	9	235	0xeb	12
32	0x20	9	83	0x53	7	134	0x86	8	185	0xb9	8	236	0xec	12
33	0x21	8	84	0x54	7	135	0x87	7	186	0xba	8	237	0xed	8
34	0x22	8	85	0x55	11	136	0x88	8	187	0xbb	9	238	0xee	7
35	0x23	12	86	0x56	8	137	0x89	8	188	0xbc	8	239	0xef	7
36	0x24	13	87	0x57	9	138	0x8a	8	189	0xbd	10	240	0xf0	9
37	0x25	7	88	0x58	7	139	0x8b	7	190	0xbe	21	241	0xf1	8
38	0x26	8	89	0x59	7	140	0x8c	12	191	0xbf	13	242	0xf2	8
39	0x27	7	90	0x5a	7	141	0x8d	9	192	0xc0	8	243	0xf3	8
40	0x28	8	91	0x5b	9	142	0x8e	14	193	0xc1	7	244	0xf4	8
41	0x29	9	92	0x5c	12	143	0x8f	10	194	0xc2	7	245	0xf5	8
42	0x2a	11	93	0x5d	9	144	0x90	9	195	0xc3	9	246	0xf6	7
43	0x2b	8	94	0x5e	9	145	0x91	33	196	0xc4	12	247	0xf7	7
44	0x2c	8	95	0x5f	10	146	0x92	10	197	0xc5	9	248	0xf8	7
45	0x2d	7	96	0x60	8	147	0x93	14	198	0xc6	8	249	0xf9	8
46	0x2e	8	97	0x61	7	148	0x94	8	199	0xc7	9	250	0xfa	8
47	0x2f	9	98	0x62	9	149	0x95	7	200	0xc8	7	251	0xfb	9
48	0x30	7	99	0x63	8	150	0x96	12	201	0xc9	7	252	0xfc	7
49	0x31	8	100	0x64	11	151	0x97	8	202	0xca	8	253	0xfd	8
50	0x32	8	101	0x65	11	152	0x98	8	203	0xcb	8	254	0xfe	7
51	0x33	7	102	0x66	8	153	0x99	8	204	0xcc	8	255	0xff	8

Table B.4: Inner c -Differential Uniformity δ_c^{-1} of Kuznyechik Inverse S-box

c	Hex	δ_c^{-1}	c	Hex	δ_c^{-1}	c	Hex	δ_c^{-1}	c	Hex	δ_c^{-1}	c	Hex	δ_c^{-1}
1	0x01	8	52	0x34	8	103	0x67	7	154	0x9a	7	205	0xcd	8
2	0x02	9	53	0x35	8	104	0x68	9	155	0x9b	7	206	0xce	7
3	0x03	8	54	0x36	8	105	0x69	7	156	0x9c	8	207	0xcf	7
4	0x04	7	55	0x37	7	106	0x6a	7	157	0x9d	7	208	0xd0	7
5	0x05	7	56	0x38	8	107	0x6b	7	158	0x9e	8	209	0xd1	8
6	0x06	7	57	0x39	8	108	0x6c	7	159	0x9f	8	210	0xd2	7
7	0x07	7	58	0x3a	8	109	0x6d	7	160	0xa0	7	211	0xd3	8
8	0x08	8	59	0x3b	8	110	0x6e	7	161	0xa1	8	212	0xd4	8
9	0x09	7	60	0x3c	7	111	0x6f	8	162	0xa2	8	213	0xd5	8
10	0x0a	8	61	0x3d	7	112	0x70	8	163	0xa3	7	214	0xd6	7
11	0x0b	8	62	0x3e	7	113	0x71	7	164	0xa4	7	215	0xd7	8
12	0x0c	7	63	0x3f	7	114	0x72	8	165	0xa5	8	216	0xd8	8
13	0x0d	8	64	0x40	9	115	0x73	8	166	0xa6	7	217	0xd9	7
14	0x0e	8	65	0x41	7	116	0x74	7	167	0xa7	7	218	0xda	7
15	0x0f	8	66	0x42	8	117	0x75	7	168	0xa8	7	219	0xdb	7
16	0x10	8	67	0x43	8	118	0x76	7	169	0xa9	8	220	0xdc	8
17	0x11	7	68	0x44	8	119	0x77	7	170	0xaa	7	221	0xdd	8
18	0x12	7	69	0x45	7	120	0x78	7	171	0xab	8	222	0xde	7
19	0x13	7	70	0x46	8	121	0x79	7	172	0xac	8	223	0xdf	7
20	0x14	8	71	0x47	7	122	0x7a	8	173	0xad	8	224	0xe0	7
21	0x15	7	72	0x48	7	123	0x7b	8	174	0xae	8	225	0xe1	9
22	0x16	8	73	0x49	8	124	0x7c	7	175	0xaf	8	226	0xe2	7
23	0x17	9	74	0x4a	8	125	0x7d	8	176	0xb0	8	227	0xe3	8

24	0x18	7	75	0x4b	7	126	0x7e	7	177	0xb1	8	228	0xe4	8
25	0x19	7	76	0x4c	7	127	0x7f	7	178	0xb2	7	229	0xe5	7
26	0x1a	8	77	0x4d	7	128	0x80	9	179	0xb3	8	230	0xe6	7
27	0x1b	9	78	0x4e	7	129	0x81	7	180	0xb4	8	231	0xe7	7
28	0x1c	8	79	0x4f	7	130	0x82	7	181	0xb5	8	232	0xe8	7
29	0x1d	8	80	0x50	7	131	0x83	8	182	0xb6	8	233	0xe9	7
30	0x1e	7	81	0x51	8	132	0x84	7	183	0xb7	7	234	0xea	7
31	0x1f	7	82	0x52	7	133	0x85	7	184	0xb8	8	235	0xeb	7
32	0x20	7	83	0x53	8	134	0x86	7	185	0xb9	8	236	0xec	7
33	0x21	7	84	0x54	8	135	0x87	8	186	0xba	7	237	0xed	8
34	0x22	8	85	0x55	8	136	0x88	7	187	0xbb	7	238	0xee	8
35	0x23	8	86	0x56	7	137	0x89	8	188	0xbc	9	239	0xef	8
36	0x24	8	87	0x57	8	138	0x8a	8	189	0xbd	8	240	0xf0	8
37	0x25	7	88	0x58	8	139	0x8b	8	190	0xbe	8	241	0xf1	8
38	0x26	8	89	0x59	8	140	0x8c	7	191	0xbf	7	242	0xf2	8
39	0x27	7	90	0x5a	8	141	0x8d	8	192	0xc0	8	243	0xf3	8
40	0x28	8	91	0x5b	8	142	0x8e	7	193	0xc1	7	244	0xf4	7
41	0x29	8	92	0x5c	7	143	0x8f	8	194	0xc2	9	245	0xf5	7
42	0x2a	8	93	0x5d	8	144	0x90	7	195	0xc3	7	246	0xf6	7
43	0x2b	8	94	0x5e	9	145	0x91	7	196	0xc4	7	247	0xf7	8
44	0x2c	7	95	0x5f	7	146	0x92	8	197	0xc5	7	248	0xf8	9
45	0x2d	8	96	0x60	7	147	0x93	7	198	0xc6	7	249	0xf9	7
46	0x2e	7	97	0x61	7	148	0x94	9	199	0xc7	8	250	0xfa	7
47	0x2f	7	98	0x62	7	149	0x95	7	200	0xc8	8	251	0xfb	8
48	0x30	7	99	0x63	8	150	0x96	7	201	0xc9	7	252	0xfc	8
49	0x31	9	100	0x64	8	151	0x97	7	202	0xca	8	253	0xfd	8
50	0x32	9	101	0x65	8	152	0x98	8	203	0xcb	8	254	0xfe	7
51	0x33	7	102	0x66	8	153	0x99	7	204	0xcc	8	255	0xff	8

Appendix C. Deterministic verification of the reduced-round characteristic searches

The reduced-round characteristic results in the main article are generated by a deterministic verifier that uses the exact Kuznyechik S-box, the RFC field representation, and the exact logarithmic transition costs. No random sampling or random seed is used in these searches. The verifier exhaustively enumerates the stated search families and includes regression assertions for the quoted optima. This section records the values reproduced by that verifier.

Table C.5: Deterministic verification of the best post- L_1 weight-one characteristics. Exponents are $-\log_2 P$.

c	2 rounds	3 rounds	4 rounds
01	89.150375	174.300750	200.810525
02	83.980120	169.715458	199.247953
e1	83.980120	169.715458	199.247953
04	89.981437	175.915699	204.063051
03	91.195485	177.345860	205.055308

For two rounds, the classical optimum occurs at $k = 4$, $\beta = 0c$ with exponent 89.1503749928, while the $c = 02/e1$ optimum occurs at $k = 2$, $\beta = 91$ with exponent 83.9801201948. For three rounds in the $16 \rightarrow 1 \rightarrow 16$

family, the corresponding exponents are 174.3007499856 and 169.7154576883, giving the 5.2-bit and 4.6-bit advantages quoted in the main article.

The post- L_1 weight-two search is also exhaustive. It checks

$$\binom{16}{2} 255^2 = 7,803,000$$

boundaries. The resulting optima are listed in Table C.6.

Table C.6: Exhaustive post- L_1 weight-two verification.

c	exponent	positions	values
01	93.320300	(4, 12)	(51, 8f)
02	85.998372	(0, 2)	(74, 82)
e1	85.998372	(0, 2)	(74, 82)
04	90.573872	(8, 12)	(5d, ed)
03	94.047951	(2, 9)	(14, 95)

For the strongest class 02/e1, the weight-two optimum $2^{-85.9983718993}$ does not improve the weight-one optimum $2^{-83.9801201948}$.

The four-round family was reconstructed independently because the older working files did not preserve the final search output. Exhaustive enumeration of the stated $16 \rightarrow 1 \rightarrow 16 \rightarrow 1$ topology gives the corrected optima $2^{-200.8105249899}$ classically and $2^{-199.2479527691}$ for $c = 02/e1$. Thus the nonclassical gain decreases to approximately 1.56 bits but remains positive in this family. These values replace the older provisional values $2^{-202.7}$ and $2^{-202.8}$.

Appendix D. Detailed cryptanalysis results by round

The following tables reproduce the highest-ranked entries printed by the exploratory analysis for each run. The column “FDR Sig.” and the marker ‡ reproduce the implementation’s Boolean flag under the thresholding rule used in that run; they should therefore be read together with the numerical corrected p -value, rather than as an assertion of significance at a universal 0.05 cutoff. The marker † records entries that were highlighted by the implementation’s data-dependent combined-evidence diagnostic. As explained in the statistical-methodology section of the main article, these historical flags are descriptive outputs of the exploratory pipeline and are not used as confirmatory evidence.

Run provenance. The recorded scans used 36 configurations. The tables headed “Seed 37” below come from the later run recorded with 128 worker processes. The tables headed “Seed 42” reproduce the historical 10-process run that contains the nine-round $c = 04$, byte-8-to-byte-8 count-130 observation. A later 256-process run carrying the same nominal seed label produced a different finite-sample outcome and is retained in the repository as a separate rerun. In the historical multiprocessing implementation, each worker initialized its Python pseudorandom generator independently from operating-system entropy using a fresh 128-bit value. Consequently, labels such as “Seed 37” and “Seed 42” identify recorded runs but do not by themselves determine all worker streams or imply bit-for-bit replayability.

Recorded run labelled Seed 37 (128 worker processes)

Table D.7: Top Noteworthy Findings for 9-Round Kuznyechik (run labelled Seed 37)

Marker	c	Configuration	Max Bias	FDR Sig.	Min FDR P-val
†	0xe1	byte_4_in→byte_4_out	1.7x	1	9.25e-03
†	0x91	byte_0_in→byte_1_out	1.5x	1	5.93e-02
†	0x91	byte_12_in→byte_12_out	1.5x	1	5.94e-02
†	0xe1	byte_12_in→byte_12_out	1.5x	1	5.94e-02
†	0x03	byte_4_in→byte_4_out	1.6x	0	1.16e-01
†	0x03	byte_6_in→byte_6_out	1.6x	0	1.22e-01
†	0x01	byte_12_in→byte_12_out	3.1x	0	2.36e-01
†	0xe1	byte_4_in→byte_5_out	1.6x	0	2.40e-01
†	0x01	byte_0_in→byte_0_out	3.0x	0	2.51e-01
†	0x91	byte_0_in→byte_0_out	1.6x	0	2.96e-01
†	0xbe	byte_14_in→byte_15_out	1.5x	0	2.97e-01
†	0xbe	byte_14_in→byte_14_out	1.5x	0	3.79e-01
†	0x02	byte_14_in→byte_15_out	1.5x	0	4.58e-01
†	0x02	byte_8_in→byte_8_out	1.5x	0	4.79e-01
†	0xe1	byte_4_in→byte_4_out	1.5x	0	5.95e-01
†	0x02	byte_12_in→byte_12_out	1.5x	0	5.96e-01
†	0x04	byte_4_in→byte_4_out	1.5x	0	6.16e-01
†	0x91	byte_8_in→byte_8_out	1.5x	0	7.14e-01
†	0x04	byte_8_in→byte_8_out	1.5x	0	7.59e-01
†	0x03	byte_12_in→byte_12_out	1.5x	0	8.28e-01

Table D.8: Top Noteworthy Findings for 8-Round Kuznyechik (run labelled Seed 37)

Marker	c	Configuration	Max Bias	FDR Sig.	Min FDR P-val
†	0x91	byte_2_in→byte_2_out	1.5x	0	2.82e-02
†	0xe1	byte_4_in→byte_4_out	1.6x	0	4.32e-02
†	0xbe	byte_14_in→byte_15_out	1.5x	0	1.22e-01
†	0x04	byte_2_in→byte_2_out	1.6x	0	1.60e-01
†	0x91	byte_12_in→byte_12_out	1.6x	0	1.83e-01
†	0x02	byte_14_in→byte_15_out	1.6x	0	1.87e-01
†	0xbe	byte_14_in→byte_14_out	1.6x	0	1.87e-01
†	0x04	byte_4_in→byte_4_out	1.5x	0	2.45e-01
†	0x03	byte_0_in→byte_0_out	1.5x	0	2.45e-01
†	0x04	byte_8_in→byte_8_out	1.5x	0	2.45e-01
†	0x02	byte_0_in→byte_1_out	1.6x	0	3.01e-01
†	0x91	byte_4_in→byte_4_out	1.6x	0	3.05e-01
†	0x03	byte_12_in→byte_12_out	1.5x	0	4.63e-01
†	0x03	byte_10_in→byte_10_out	1.5x	0	4.78e-01
†	0xe1	byte_6_in→byte_6_out	1.6x	0	4.79e-01
†	0xbe	byte_2_in→byte_2_out	1.5x	0	5.93e-01
†	0xe1	byte_12_in→byte_12_out	1.5x	0	6.22e-01
†	0xbe	byte_6_in→byte_6_out	1.5x	0	7.18e-01
†	0xbe	byte_0_in→byte_0_out	1.5x	0	7.51e-01
†	0xe1	byte_4_in→byte_5_out	1.5x	0	7.58e-01

Table D.9: Top Noteworthy Findings for 7-Round Kuznyechik (run labelled Seed 37)

Marker	c	Configuration	Max Bias	FDR Sig.	Min FDR P-val
†	0x03	byte_10_in→byte_10_out	1.6x	0	1.29e-02
†	0x04	byte_8_in→byte_8_out	1.6x	0	2.60e-02
†	0xe1	byte_4_in→byte_5_out	1.6x	0	5.93e-02
†	0xbe	byte_6_in→byte_6_out	1.6x	0	7.10e-02
†	0x02	byte_8_in→byte_8_out	1.6x	0	2.40e-01
†	0x02	byte_12_in→byte_12_out	1.6x	0	3.01e-01
†	0x03	byte_12_in→byte_12_out	1.6x	0	3.01e-01
†	0x03	byte_0_in→byte_0_out	1.6x	0	4.79e-01
†	0xbe	byte_2_in→byte_2_out	1.5x	0	4.79e-01
†	0xbe	byte_14_in→byte_15_out	1.5x	0	4.79e-01
†	0xe1	byte_8_in→byte_8_out	1.5x	0	5.94e-01

Table D.9 – continued from previous page

Marker	c	Configuration	Max Bias	FDR Sig.	Min FDR P-val
†	0x91	byte_0_in→byte_1_out	1.6x	0	7.12e-01
†	0x01	byte_6_in→byte_6_out	3.0x	0	7.12e-01
†	0x01	byte_0_in→byte_0_out	3.0x	0	7.12e-01
†	0xbe	byte_0_in→byte_0_out	1.5x	0	7.14e-01
†	0x91	byte_0_in→byte_0_out	1.5x	0	7.28e-01
†	0x04	byte_4_in→byte_4_out	1.5x	0	7.57e-01
†	0x04	byte_2_in→byte_2_out	1.5x	0	7.78e-01
†	0x02	byte_14_in→byte_15_out	1.5x	0	7.92e-01
†	0x03	byte_6_in→byte_6_out	1.5x	0	7.98e-01

Table D.10: Top Noteworthy Findings for 6-Round Kuznyechik (run labelled Seed 37)

Marker	c	Configuration	Max Bias	FDR Sig.	Min FDR P-val
†	0xe1	byte_6_in→byte_6_out	1.5x	0	2.82e-02
†	0x03	byte_4_in→byte_4_out	1.6x	0	4.31e-02
†	0x02	byte_12_in→byte_12_out	1.5x	0	1.22e-01
†	0x91	byte_8_in→byte_8_out	1.6x	0	1.51e-01
†	0x91	byte_0_in→byte_1_out	1.5x	0	2.45e-01
†	0xbe	byte_0_in→byte_0_out	1.5x	0	2.45e-01
†	0xbe	byte_2_in→byte_2_out	1.5x	0	2.45e-01
†	0x91	byte_0_in→byte_0_out	1.5x	0	3.79e-01
†	0xbe	byte_6_in→byte_6_out	1.5x	0	3.96e-01
†	0x04	byte_2_in→byte_2_out	1.5x	0	4.27e-01
†	0x91	byte_12_in→byte_12_out	1.5x	0	4.57e-01
†	0x02	byte_14_in→byte_15_out	1.6x	0	4.79e-01
†	0xbe	byte_14_in→byte_15_out	1.5x	0	5.19e-01
†	0x91	byte_4_in→byte_4_out	1.5x	0	5.94e-01
†	0xe1	byte_4_in→byte_5_out	1.5x	0	6.16e-01
†	0xe1	byte_8_in→byte_8_out	1.5x	0	6.16e-01
†	0xbe	byte_14_in→byte_14_out	1.5x	0	6.16e-01
†	0x04	byte_8_in→byte_8_out	1.5x	0	6.16e-01
†	0x03	byte_10_in→byte_10_out	1.5x	0	7.67e-01
†	0x02	byte_2_in→byte_2_out	1.5x	0	7.77e-01

Table D.11: Top Noteworthy Findings for 5-Round Kuznyechik (run labelled Seed 37)

Marker	c	Configuration	Max Bias	FDR Sig.	Min FDR P-val
†	0x04	byte_4_in→byte_4_out	1.6x	0	1.16e-01
†	0x03	byte_6_in→byte_6_out	1.5x	0	1.22e-01
†	0xe1	byte_8_in→byte_8_out	1.5x	0	1.22e-01
†	0x03	byte_10_in→byte_10_out	1.6x	0	1.51e-01
†	0xbe	byte_14_in→byte_14_out	1.6x	0	1.87e-01
†	0x91	byte_4_in→byte_4_out	1.5x	0	2.40e-01
†	0x02	byte_8_in→byte_8_out	1.5x	0	2.45e-01
†	0x02	byte_12_in→byte_12_out	1.6x	0	2.97e-01
†	0x01	byte_2_in→byte_2_out	3.1x	0	3.56e-01
†	0x01	byte_6_in→byte_6_out	3.0x	0	3.56e-01
†	0x91	byte_12_in→byte_12_out	1.5x	0	4.27e-01
†	0x02	byte_0_in→byte_1_out	1.5x	0	4.44e-01
†	0x04	byte_8_in→byte_8_out	1.5x	0	4.58e-01
†	0xbe	byte_6_in→byte_6_out	1.6x	0	4.78e-01
†	0x01	byte_12_in→byte_12_out	3.1x	0	4.79e-01
†	0x01	byte_8_in→byte_8_out	3.1x	0	4.79e-01
†	0x91	byte_2_in→byte_2_out	1.5x	0	4.79e-01
†	0x02	byte_14_in→byte_15_out	1.5x	0	5.93e-01
†	0xbe	byte_14_in→byte_15_out	1.5x	0	5.94e-01
†	0x02	byte_2_in→byte_2_out	1.5x	0	5.94e-01

Table D.12: Top Noteworthy Findings for 4-Round Kuznyechik (run labelled Seed 37)

Marker	c	Configuration	Max Bias	FDR Sig.	Min FDR P-val
†	0xe1	byte_4_in→byte_4_out	1.7x	0	9.25e-03
†	0x91	byte_0_in→byte_1_out	1.6x	0	2.59e-02
†	0x03	byte_0_in→byte_0_out	1.5x	0	5.94e-02
†	0x03	byte_4_in→byte_4_out	1.6x	0	1.16e-01
†	0x04	byte_8_in→byte_8_out	1.6x	0	1.16e-01
†	0x02	byte_12_in→byte_12_out	1.5x	0	1.22e-01
†	0x01	byte_2_in→byte_2_out	3.1x	0	2.24e-01
†	0x91	byte_12_in→byte_12_out	1.5x	0	2.39e-01
†	0x02	byte_2_in→byte_2_out	1.6x	0	2.40e-01
†	0xe1	byte_8_in→byte_8_out	1.5x	0	2.45e-01
†	0x02	byte_14_in→byte_15_out	1.6x	0	3.01e-01
†	0x91	byte_8_in→byte_8_out	1.6x	0	3.01e-01
†	0xe1	byte_6_in→byte_6_out	1.6x	0	3.01e-01
†	0xbe	byte_0_in→byte_0_out	1.5x	0	3.41e-01
†	0x04	byte_2_in→byte_2_out	1.6x	0	4.57e-01
†	0xbe	byte_6_in→byte_6_out	1.6x	0	4.78e-01
†	0x03	byte_6_in→byte_6_out	1.6x	0	4.79e-01
†	0xbe	byte_14_in→byte_15_out	1.5x	0	4.79e-01
†	0x04	byte_4_in→byte_4_out	1.5x	0	4.80e-01
†	0x91	byte_2_in→byte_2_out	1.5x	0	5.25e-01

Table D.13: Top Noteworthy Findings for 3-Round Kuznyechik (run labelled Seed 37)

Marker	c	Configuration	Max Bias	FDR Sig.	Min FDR P-val
†	0x01	byte_0_in→byte_0_out	9.8x	10883	0.00e+00
†	0x01	byte_12_in→byte_12_out	10.1x	11006	0.00e+00
†	0x01	byte_2_in→byte_2_out	10.7x	11047	0.00e+00
†	0x01	byte_2_in→byte_3_out	10.3x	10984	0.00e+00
†	0x01	byte_6_in→byte_6_out	10.6x	10958	0.00e+00
†	0x01	byte_8_in→byte_8_out	9.9x	10920	0.00e+00
†	0x03	byte_4_in→byte_4_out	1.5x	1	1.07e-03
†	0xbe	byte_14_in→byte_14_out	1.6x	0	2.60e-02
†	0xe1	byte_12_in→byte_12_out	1.5x	0	2.81e-02
†	0x01	byte_8_in→byte_8_out	3.3x	0	6.73e-02
†	0xbe	byte_6_in→byte_6_out	1.6x	0	7.09e-02
†	0x02	byte_2_in→byte_2_out	1.5x	0	1.22e-01
†	0xe1	byte_2_in→byte_2_out	1.6x	0	1.87e-01
†	0x04	byte_8_in→byte_8_out	1.6x	0	2.40e-01
†	0x91	byte_8_in→byte_8_out	1.6x	0	2.40e-01
†	0x03	byte_10_in→byte_10_out	1.5x	0	2.45e-01
†	0x03	byte_12_in→byte_12_out	1.6x	0	3.01e-01
†	0x03	byte_0_in→byte_0_out	1.5x	0	3.78e-01
†	0xe1	byte_4_in→byte_5_out	1.6x	0	4.58e-01
†	0x02	byte_8_in→byte_8_out	1.5x	0	4.58e-01

Table D.14: Top Noteworthy Findings for 2-Round Kuznyechik (run labelled Seed 37)

Marker	c	Configuration	Max Bias	FDR Sig.	Min FDR P-val
†	0x91	byte_8_in→byte_8_out	1.7x	0	1.56e-02
†	0x03	byte_12_in→byte_12_out	1.6x	0	5.94e-02
†	0x04	byte_4_in→byte_4_out	1.6x	0	1.22e-01
†	0x03	byte_10_in→byte_10_out	1.6x	0	1.87e-01
†	0xe1	byte_4_in→byte_5_out	1.5x	0	2.45e-01
†	0xbe	byte_14_in→byte_14_out	1.6x	0	4.44e-01
†	0xbe	byte_6_in→byte_6_out	1.5x	0	4.57e-01
†	0xe1	byte_2_in→byte_2_out	1.5x	0	4.79e-01
†	0x91	byte_0_in→byte_1_out	1.6x	0	4.79e-01
†	0xbe	byte_14_in→byte_15_out	1.5x	0	5.69e-01
†	0x02	byte_12_in→byte_12_out	1.5x	0	5.87e-01

Table D.14 – continued from previous page

Marker	c	Configuration	Max Bias	FDR Sig.	Min FDR P-val
†	0x02	byte_8_in→byte_8_out	1.5x	0	5.93e-01
†	0xe1	byte_8_in→byte_8_out	1.5x	0	7.18e-01
†	0x01	byte_2_in→byte_2_out	2.9x	0	7.52e-01
†	0x01	byte_6_in→byte_6_out	2.9x	0	7.52e-01
†	0xbe	byte_2_in→byte_2_out	1.5x	0	7.78e-01
†	0x02	byte_14_in→byte_15_out	1.5x	0	7.78e-01
†	0x03	byte_0_in→byte_0_out	1.5x	0	9.08e-01
†	0x02	byte_2_in→byte_2_out	1.5x	0	9.09e-01
†	0x91	byte_0_in→byte_0_out	1.5x	0	9.25e-01

Table D.15: Top Noteworthy Findings for 1-Round Kuznyechik (run labelled Seed 37)

Marker	c	Configuration	Max Bias	FDR Sig.	Min FDR P-val
‡	0x01	byte_0_in→byte_0_out	9.5x	10909	0.00e+00
‡	0x01	byte_12_in→byte_12_out	10.2x	10990	0.00e+00
‡	0x01	byte_2_in→byte_2_out	10.6x	10989	0.00e+00
‡	0x01	byte_2_in→byte_3_out	10.2x	10923	0.00e+00
‡	0x01	byte_6_in→byte_6_out	10.7x	11038	0.00e+00
‡	0x01	byte_8_in→byte_8_out	9.7x	10941	0.00e+00
†	0x91	byte_8_in→byte_8_out	1.7x	0	1.56e-02
†	0x03	byte_12_in→byte_12_out	1.6x	0	5.94e-02
†	0x04	byte_4_in→byte_4_out	1.6x	0	1.22e-01
†	0x03	byte_10_in→byte_10_out	1.6x	0	1.87e-01
†	0xe1	byte_4_in→byte_5_out	1.5x	0	2.45e-01
†	0xbe	byte_14_in→byte_14_out	1.6x	0	4.44e-01
†	0xbe	byte_6_in→byte_6_out	1.5x	0	4.57e-01
†	0xe1	byte_2_in→byte_2_out	1.5x	0	4.79e-01
†	0x91	byte_0_in→byte_1_out	1.6x	0	4.79e-01
†	0xbe	byte_14_in→byte_15_out	1.5x	0	5.69e-01
†	0x02	byte_12_in→byte_12_out	1.5x	0	5.87e-01
†	0x02	byte_8_in→byte_8_out	1.5x	0	5.93e-01
†	0xe1	byte_8_in→byte_8_out	1.5x	0	7.18e-01
†	0x01	byte_2_in→byte_2_out	2.9x	0	7.52e-01

Historical run labelled Seed 42 (10 worker processes)

Table D.16: Top Noteworthy Findings for 9-Round Kuznyechik (historical run labelled Seed 42)

Marker	c	Configuration	Max Bias	FDR Sig.	Min FDR P-val
‡	0x04	byte_8_in→byte_8_out	1.7x	1	1.85e-03
‡	0xe1	byte_6_in→byte_6_out	1.7x	1	9.24e-03
†	0x02	byte_12_in→byte_12_out	1.6x	0	7.08e-02
†	0x02	byte_0_in→byte_1_out	1.6x	0	1.22e-01
†	0xbe	byte_14_in→byte_14_out	1.6x	0	1.51e-01
†	0x03	byte_12_in→byte_12_out	1.6x	0	1.87e-01
†	0xbe	byte_2_in→byte_2_out	1.5x	0	1.90e-01
†	0x03	byte_6_in→byte_6_out	1.5x	0	2.40e-01
†	0xe1	byte_8_in→byte_8_out	1.6x	0	2.40e-01
†	0xe1	byte_4_in→byte_4_out	1.5x	0	2.45e-01
†	0xbe	byte_14_in→byte_15_out	1.5x	0	3.79e-01
†	0x91	byte_12_in→byte_12_out	1.5x	0	4.58e-01
†	0x04	byte_4_in→byte_4_out	1.5x	0	4.63e-01
†	0x02	byte_2_in→byte_2_out	1.5x	0	4.63e-01
†	0x04	byte_2_in→byte_2_out	1.6x	0	4.79e-01
†	0x01	byte_2_in→byte_3_out	3.0x	0	7.12e-01
†	0x01	byte_0_in→byte_0_out	3.0x	0	7.12e-01
†	0x01	byte_6_in→byte_6_out	2.9x	0	7.12e-01
†	0x01	byte_12_in→byte_12_out	2.9x	0	7.12e-01
†	0x01	byte_2_in→byte_2_out	2.9x	0	7.12e-01

Table D.17: Top Noteworthy Findings for 8-Round Kuznyechik (historical run labelled Seed 42)

Marker	c	Configuration	Max Bias	FDR Sig.	Min FDR P-val
†	0x03	byte_12_in→byte_12_out	1.6x	1	5.81e-03
†	0x03	byte_4_in→byte_4_out	1.6x	0	4.31e-02
†	0x91	byte_2_in→byte_2_out	1.5x	0	5.93e-02
†	0xe1	byte_6_in→byte_6_out	1.6x	0	7.09e-02
†	0xe1	byte_4_in→byte_4_out	1.5x	0	1.22e-01
†	0x04	byte_8_in→byte_8_out	1.5x	0	2.40e-01
†	0x04	byte_4_in→byte_4_out	1.5x	0	2.45e-01
†	0x02	byte_0_in→byte_1_out	1.5x	0	2.45e-01
†	0x03	byte_6_in→byte_6_out	1.5x	0	4.58e-01
†	0xbe	byte_14_in→byte_14_out	1.6x	0	4.80e-01
†	0x91	byte_0_in→byte_1_out	1.5x	0	5.87e-01
†	0xbe	byte_14_in→byte_15_out	1.5x	0	5.95e-01
†	0x01	byte_0_in→byte_0_out	2.9x	0	1.00e+00
†	0x01	byte_12_in→byte_12_out	2.9x	0	1.00e+00
†	0x01	byte_2_in→byte_2_out	2.9x	0	1.00e+00
†	0x01	byte_2_in→byte_3_out	2.9x	0	1.00e+00
†	0x01	byte_6_in→byte_6_out	2.7x	0	1.00e+00
†	0x01	byte_8_in→byte_8_out	2.9x	0	1.00e+00
†	0x02	byte_12_in→byte_12_out	1.5x	0	1.00e+00
†	0x02	byte_14_in→byte_15_out	1.5x	0	1.00e+00

Table D.18: Top Noteworthy Findings for 7-Round Kuznyechik (historical run labelled Seed 42)

Marker	c	Configuration	Max Bias	FDR Sig.	Min FDR P-val
†	0x03	byte_6_in→byte_6_out	1.6x	0	1.16e-01
†	0x91	byte_2_in→byte_2_out	1.6x	0	1.16e-01
†	0x02	byte_8_in→byte_8_out	1.5x	0	1.22e-01
†	0xe1	byte_4_in→byte_5_out	1.6x	0	1.50e-01
†	0xbe	byte_2_in→byte_2_out	1.6x	0	1.87e-01
†	0xe1	byte_4_in→byte_4_out	1.6x	0	2.40e-01
†	0xbe	byte_0_in→byte_0_out	1.5x	0	2.45e-01
†	0x04	byte_4_in→byte_4_out	1.5x	0	2.45e-01
†	0x02	byte_0_in→byte_1_out	1.6x	0	3.00e-01
†	0xbe	byte_6_in→byte_6_out	1.6x	0	3.01e-01
†	0x01	byte_0_in→byte_0_out	2.9x	0	1.00e+00
†	0x01	byte_12_in→byte_12_out	2.7x	0	1.00e+00
†	0x01	byte_2_in→byte_2_out	2.7x	0	1.00e+00
†	0x01	byte_2_in→byte_3_out	2.9x	0	1.00e+00
†	0x01	byte_6_in→byte_6_out	2.6x	0	1.00e+00
†	0x01	byte_8_in→byte_8_out	2.7x	0	1.00e+00
†	0x02	byte_12_in→byte_12_out	1.5x	0	1.00e+00
†	0x02	byte_14_in→byte_15_out	1.5x	0	1.00e+00
†	0x02	byte_2_in→byte_2_out	1.5x	0	1.00e+00
†	0x03	byte_0_in→byte_0_out	1.5x	0	1.00e+00

Table D.19: Top Noteworthy Findings for 6-Round Kuznyechik (historical run labelled Seed 42)

Marker	c	Configuration	Max Bias	FDR Sig.	Min FDR P-val
†	0x02	byte_14_in→byte_15_out	1.6x	0	4.31e-02
†	0xbe	byte_2_in→byte_2_out	1.6x	0	1.16e-01
†	0xe1	byte_6_in→byte_6_out	1.5x	0	1.22e-01
†	0xe1	byte_4_in→byte_5_out	1.5x	0	1.22e-01
†	0x02	byte_12_in→byte_12_out	1.6x	0	2.97e-01
†	0x02	byte_2_in→byte_2_out	1.6x	0	3.00e-01
†	0x02	byte_8_in→byte_8_out	1.6x	0	3.01e-01
†	0x91	byte_0_in→byte_1_out	1.6x	0	3.01e-01
†	0x01	byte_2_in→byte_3_out	3.0x	0	3.56e-01
†	0x03	byte_12_in→byte_12_out	1.6x	0	3.70e-01
†	0x01	byte_0_in→byte_0_out	2.9x	0	1.00e+00

Table D.19 – continued from previous page

Marker	c	Configuration	Max Bias	FDR Sig.	Min FDR P-val
†	0x01	byte_12_in→byte_12_out	2.7x	0	1.00e+00
†	0x01	byte_2_in→byte_2_out	2.9x	0	1.00e+00
†	0x01	byte_6_in→byte_6_out	2.7x	0	1.00e+00
†	0x01	byte_8_in→byte_8_out	2.7x	0	1.00e+00
†	0x02	byte_0_in→byte_1_out	1.5x	0	1.00e+00
†	0x03	byte_0_in→byte_0_out	1.5x	0	1.00e+00
†	0x03	byte_10_in→byte_10_out	1.5x	0	1.00e+00
†	0x03	byte_4_in→byte_4_out	1.5x	0	1.00e+00
†	0x03	byte_6_in→byte_6_out	1.5x	0	1.00e+00

Table D.20: Top Noteworthy Findings for 5-Round Kuznyechik (historical run labelled Seed 42)

Marker	c	Configuration	Max Bias	FDR Sig.	Min FDR P-val
†	0x03	byte_12_in→byte_12_out	1.6x	0	9.36e-02
†	0x04	byte_8_in→byte_8_out	1.6x	0	9.37e-02
†	0x91	byte_12_in→byte_12_out	1.5x	0	1.22e-01
†	0x02	byte_8_in→byte_8_out	1.6x	0	1.50e-01
†	0x01	byte_2_in→byte_3_out	3.1x	0	2.24e-01
†	0xe1	byte_4_in→byte_5_out	1.5x	0	2.40e-01
†	0x03	byte_0_in→byte_0_out	1.5x	0	2.45e-01
†	0x02	byte_14_in→byte_15_out	1.6x	0	3.05e-01
†	0xe1	byte_2_in→byte_2_out	1.5x	0	3.78e-01
†	0x91	byte_0_in→byte_1_out	1.5x	0	3.96e-01
†	0x01	byte_0_in→byte_0_out	2.9x	0	1.00e+00
†	0x01	byte_12_in→byte_12_out	2.9x	0	1.00e+00
†	0x01	byte_2_in→byte_2_out	2.9x	0	1.00e+00
†	0x01	byte_6_in→byte_6_out	2.6x	0	1.00e+00
†	0x01	byte_8_in→byte_8_out	2.9x	0	1.00e+00
†	0x02	byte_0_in→byte_1_out	1.5x	0	1.00e+00
†	0x02	byte_12_in→byte_12_out	1.5x	0	1.00e+00
†	0x02	byte_2_in→byte_2_out	1.5x	0	1.00e+00
†	0x03	byte_10_in→byte_10_out	1.5x	0	1.00e+00
†	0x03	byte_4_in→byte_4_out	1.5x	0	1.00e+00

Table D.21: Top Noteworthy Findings for 4-Round Kuznyechik (historical run labelled Seed 42)

Marker	c	Configuration	Max Bias	FDR Sig.	Min FDR P-val
†	0x91	byte_8_in→byte_8_out	1.7x	0	1.56e-02
†	0x03	byte_12_in→byte_12_out	1.6x	0	5.94e-02
†	0x04	byte_4_in→byte_4_out	1.6x	0	1.22e-01
†	0x03	byte_10_in→byte_10_out	1.6x	0	1.87e-01
†	0xe1	byte_4_in→byte_5_out	1.5x	0	2.45e-01
†	0xbe	byte_14_in→byte_14_out	1.6x	0	4.44e-01
†	0xbe	byte_6_in→byte_6_out	1.5x	0	4.57e-01
†	0xe1	byte_2_in→byte_2_out	1.5x	0	4.79e-01
†	0x91	byte_0_in→byte_1_out	1.6x	0	4.79e-01
†	0xbe	byte_14_in→byte_15_out	1.5x	0	5.69e-01
†	0x01	byte_0_in→byte_0_out	2.9x	0	1.00e+00
†	0x01	byte_12_in→byte_12_out	2.9x	0	1.00e+00
†	0x01	byte_2_in→byte_2_out	2.9x	0	1.00e+00
†	0x01	byte_2_in→byte_3_out	2.7x	0	1.00e+00
†	0x01	byte_6_in→byte_6_out	2.9x	0	1.00e+00
†	0x01	byte_8_in→byte_8_out	2.9x	0	1.00e+00
†	0x02	byte_0_in→byte_1_out	1.5x	0	1.00e+00
†	0x02	byte_12_in→byte_12_out	1.5x	0	1.00e+00
†	0x02	byte_14_in→byte_15_out	1.5x	0	1.00e+00
†	0x02	byte_2_in→byte_2_out	1.5x	0	1.00e+00

Table D.22: Top Noteworthy Findings for 3-Round Kuznyechik (historical run labelled Seed 42)

Marker	c	Configuration	Max Bias	FDR Sig.	Min FDR P-val
†	0x01	byte_0_in→byte_0_out	10.1x	10896	0.00e+00
†	0x01	byte_12_in→byte_12_out	10.3x	10996	0.00e+00
†	0x01	byte_2_in→byte_2_out	10.0x	10955	0.00e+00
†	0x01	byte_2_in→byte_3_out	10.1x	10985	0.00e+00
†	0x01	byte_6_in→byte_6_out	10.4x	10991	0.00e+00
†	0x01	byte_8_in→byte_8_out	10.4x	10980	0.00e+00
†	0x02	byte_12_in→byte_12_out	1.6x	0	1.51e-01
†	0x04	byte_8_in→byte_8_out	1.5x	0	1.60e-01
†	0x91	byte_12_in→byte_12_out	1.5x	0	2.45e-01
†	0xbe	byte_14_in→byte_14_out	1.5x	0	3.01e-01
†	0x91	byte_2_in→byte_2_out	1.5x	0	3.01e-01
†	0xe1	byte_12_in→byte_12_out	1.5x	0	3.01e-01
†	0x91	byte_4_in→byte_4_out	1.5x	0	3.79e-01
†	0x02	byte_2_in→byte_2_out	1.5x	0	4.78e-01
†	0xbe	byte_2_in→byte_2_out	1.5x	0	4.80e-01
†	0x03	byte_0_in→byte_0_out	1.5x	0	5.95e-01
†	0x04	byte_2_in→byte_2_out	1.5x	0	5.95e-01
†	0xe1	byte_4_in→byte_5_out	1.5x	0	6.16e-01
†	0x03	byte_10_in→byte_10_out	1.5x	0	7.12e-01
†	0x02	byte_14_in→byte_15_out	1.5x	0	7.13e-01

Table D.23: Top Noteworthy Findings for 2-Round Kuznyechik (historical run labelled Seed 42)

Marker	c	Configuration	Max Bias	FDR Sig.	Min FDR P-val
†	0xbe	byte_6_in→byte_6_out	1.7x	0	2.81e-02
†	0x02	byte_0_in→byte_1_out	1.6x	0	4.31e-02
†	0xe1	byte_6_in→byte_6_out	1.6x	0	5.93e-02
†	0x03	byte_10_in→byte_10_out	1.6x	0	7.07e-02
†	0x03	byte_4_in→byte_4_out	1.5x	0	1.22e-01
†	0x02	byte_8_in→byte_8_out	1.5x	0	1.87e-01
†	0x04	byte_8_in→byte_8_out	1.5x	0	2.45e-01
†	0x02	byte_12_in→byte_12_out	1.5x	0	2.97e-01
†	0x91	byte_2_in→byte_2_out	1.5x	0	3.01e-01
†	0x03	byte_6_in→byte_6_out	1.5x	0	4.57e-01
†	0xbe	byte_14_in→byte_14_out	1.5x	0	4.58e-01
†	0xe1	byte_8_in→byte_8_out	1.5x	0	4.79e-01
†	0xbe	byte_0_in→byte_0_out	1.5x	0	4.79e-01
†	0x91	byte_0_in→byte_0_out	1.5x	0	4.80e-01
†	0xbe	byte_2_in→byte_2_out	1.5x	0	8.30e-01
†	0x91	byte_4_in→byte_4_out	1.5x	0	8.54e-01
†	0xbe	byte_14_in→byte_15_out	1.5x	0	9.95e-01
†	0x01	byte_12_in→byte_12_out	2.9x	0	1.00e+00
†	0x01	byte_2_in→byte_2_out	2.7x	0	1.00e+00
†	0x01	byte_2_in→byte_3_out	2.9x	0	1.00e+00

Table D.24: Top Noteworthy Findings for 1-Round Kuznyechik (historical run labelled Seed 42)

Marker	c	Configuration	Max Bias	FDR Sig.	Min FDR P-val
†	0x01	byte_0_in→byte_0_out	10.3x	10959	0.00e+00
†	0x01	byte_12_in→byte_12_out	10.1x	11029	0.00e+00
†	0x01	byte_2_in→byte_2_out	10.1x	10976	0.00e+00
†	0x01	byte_2_in→byte_3_out	10.3x	10960	0.00e+00
†	0x01	byte_6_in→byte_6_out	10.0x	10986	0.00e+00
†	0x01	byte_8_in→byte_8_out	10.2x	10972	0.00e+00
†	0xbe	byte_6_in→byte_6_out	1.7x	0	2.81e-02
†	0x02	byte_0_in→byte_1_out	1.6x	0	4.31e-02
†	0xe1	byte_6_in→byte_6_out	1.6x	0	5.93e-02
†	0x03	byte_10_in→byte_10_out	1.6x	0	7.07e-02
†	0x03	byte_4_in→byte_4_out	1.5x	0	1.22e-01

0x00000000000000970000000000000000 -> 0x00000000000000c60000000000000000 (Bias: 1.26x, p: 0.033)
0x0000000000000009e000000000000000 -> 0x00000000000000540000000000000000 (Bias: 1.32x, p: 0.008)

SIGNIFICANT DIFFERENTIAL PAIRS (FDR-corrected $p < 7.033e-02$):

(These are specific (Input Diff, Output Diff) pairs whose observed frequencies are statistically different from expected, after multiple-test correction.)

Found 1 significant pairs.

Top 10 by corrected p-value:

Input Diff (A)	Output Diff (B)	Obs Count	Bias	Corr P-val
0x00000000000000290000000000000000	0x000000000000008d0000000000000000	130	1.7x	1.85e-03

Historical program diagnostic: COMBINED SIGNIFICANCE DETECTED for 9 rounds..

Config: byte_8_in->byte_8_out, c=0x4

Found 342 pairs with bias > 1.3 AND $p < 0.1$

- 0x00000000000000290000000000000000 -> 0x000000000000008d0000000000000000 (Bias: 1.70x, p: 0.000)
- 0x00000000000000d30000000000000000 -> 0x00000000000000540000000000000000 (Bias: 1.52x, p: 0.000)
- 0x00000000000000e40000000000000000 -> 0x00000000000000480000000000000000 (Bias: 1.51x, p: 0.000)

Historical program diagnostic: BIAS PERSISTENCE ANOMALY for 9 rounds..

Config: byte_8_in->byte_8_out, c=0x4

Observed bias: 1.70x vs Expected: 0.125x

Ratio: 13.6x higher than expected decay

Historical program message (verbatim): CRITICAL ALERT: Statistically significant characteristic found for 9 rounds..

Config: byte_8_in->byte_8_out, c=0x4

Found 1 significant pairs (threshold: 1): Input->Output

0x00000000000000290000000000000000 -> 0x000000000000008d0000000000000000 (Bias:1.7, p-val:1.85e-03)

References

- [1] E. Biham and A. Shamir, “Differential cryptanalysis of DES-like cryptosystems,” *Journal of Cryptology*, vol. 4, no. 1, pp. 3–72, 1991.
- [2] E. Biham and A. Shamir, “Differential cryptanalysis of the full 16-round DES,” in *Advances in Cryptology—CRYPTO ’92*, LNCS 740, pp. 487–496, Springer, 1993.
- [3] L. R. Knudsen, “Truncated and higher order differentials,” in *Fast Software Encryption—FSE 1994*, LNCS 1008, pp. 196–211, Springer, 1995.
- [4] P. Ellingsen, P. Felke, C. Riera, P. Stănică, and A. Tkachenko, “ c -Differentials, multiplicative uniformity, and (almost) perfect c -nonlinearity,” *IEEE Transactions on Information Theory*, vol. 66, no. 9, pp. 5781–5789, 2020.
- [5] N. Borisov, M. Chew, R. Johnson, and D. Wagner, “Multiplicative differentials,” in *Fast Software Encryption—FSE 2002*, LNCS 2365, pp. 17–33, Springer, 2002.
- [6] D. Wagner, “Towards a unifying view of block cipher cryptanalysis,” in *Fast Software Encryption—FSE 2004*, LNCS 3017, pp. 16–33, Springer, 2004.

- [7] A. Biryukov, L. Perrin, and A. Udovenko, “Reverse-engineering the S-box of Streebog, Kuznyechik and STRIBOBr1,” in *Advances in Cryptology—EUROCRYPT 2016*, LNCS 9665, pp. 372–402, Springer, 2016.
- [8] L. Perrin and A. Udovenko, “Exponential S-boxes: a link between the S-boxes of BelT and Kuznyechik/Streebog,” *IACR Transactions on Symmetric Cryptology*, 2016(2), pp. 99–124, 2017.
- [9] L. Perrin, “Partitions in the S-box of Streebog and Kuznyechik,” *IACR Transactions on Symmetric Cryptology*, 2019(1), pp. 302–329, 2019.
- [10] R. Cheng, Y. Zhou, X. Dong, and X. Du, “On the sum-of-squares of differential distribution table for (n, n) -functions,” *IEICE Transactions on Fundamentals*, vol. E105.A, no. 9, pp. 1322–1329, 2022.
- [11] S. Eddahmani and S. Mesnager, “The c -differential-linear connectivity table of vectorial Boolean functions,” *Entropy*, vol. 26, no. 3, art. 188, 2024.
- [12] J. Baudrin, P. Felke, G. Leander, P. Neumann, L. Perrin, and L. Stennes, “Commutative cryptanalysis made practical,” *IACR Transactions on Symmetric Cryptology*, 2023(4), pp. 299–329, 2023.
- [13] J. Baudrin, C. Beierle, P. Felke, G. Leander, P. Neumann, L. Perrin, and L. Stennes, “Commutative cryptanalysis as a generalization of differential cryptanalysis,” *Designs, Codes and Cryptography*, vol. 93, pp. 3243–3281, 2025.
- [14] D. Bartoli, L. Kölsch, and G. Micheli, “Differential biases, c -differential uniformity, and their relation to differential attacks,” in *WAIFI 2024 Revised Selected Papers*, LNCS 15176, pp. 191–212, Springer, 2025.
- [15] T. Peyrin and Q. Q. Tan, “Mind your path: on (key) dependencies in differential characteristics,” *IACR Transactions on Symmetric Cryptology*, 2022(4), pp. 179–207, 2022.
- [16] T. Beyne and V. Rijmen, “Differential cryptanalysis in the fixed-key model,” in *Advances in Cryptology—CRYPTO 2022, Part III*, LNCS 13509, pp. 687–716, Springer, 2022.
- [17] T. Peyrin, Q. Q. Tan, H. Zhang, and C. Zhou, “Trail-Estimator: an automated verifier for differential trails in block ciphers,” *IACR Transactions on Symmetric Cryptology*, 2025(3), pp. 475–515, 2025.
- [18] M. Nageler, S. Ghosh, M. Jüttler, and M. Eichlseder, “AutoDiVer: automatically verifying differential characteristics and learning key conditions,” *IACR Transactions on Symmetric Cryptology*, 2025(1), pp. 471–514, 2025.
- [19] P. Stănică, “Supplementary implementation, computation logs, and reproducibility code for the Kuznyechik inner- c study,” GitHub repository, 2026. [Online]. Available: <https://github.com/pstanica/KuznyechikTruncate>

d_cDU.

- [20] V. A. Kiryukhin, “Exact maximum expected differential and linear probability for 2-round Kuznyechik,” IACR Cryptology ePrint Archive, Report 2018/1085, 2018.
- [21] V. A. Kiryukhin, “Exact maximum expected differential and linear probability for 2-round Kuznyechik,” *Mathematical Aspects of Cryptography*, vol. 10, no. 2, pp. 107–116, 2019.
- [22] V. A. Kiryukhin, “An algorithm for computing the upper bound for non-minimum weight differentials in 2-round LSX-ciphers,” *Mathematical Aspects of Cryptography*, vol. 12, no. 2, pp. 93–109, 2021, doi:10.4213/mvk358.
- [23] S. Rønjom, A. Sandrib, and J. Sunde, “Subspace differential uniformity,” *Cryptography and Communications*, 2026.
- [24] R. AlTawy and A. M. Youssef, “A meet in the middle attack on reduced round Kuznyechik,” *IEICE Transactions on Fundamentals*, vol. E98.A, no. 10, pp. 2194–2198, 2015.
- [25] E. Ishchukova, E. Tolomanenko, and L. K. Babenko, “Differential analysis of 3 round Kuznyechik,” in *Proceedings of SIN 2017*, pp. 29–36, ACM, 2017.
- [26] A. Geary, M. Calderini, C. Riera, and P. Stănică, “Higher order c -differentials,” in *Security and Privacy, ICSP 2021*, CCIS 1497, pp. 3–15, Springer, 2021.